ASSIGNMENT-9

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QUESTION:

Interacting with DeFi Protocols (Testnet)

Objective

To interact with a decentralized finance (DeFi) protocol on a blockchain testnet by writing and executing a smart contract that deposits and withdraws tokens, demonstrating basic DeFi operations without real funds.

Requirements

• Install Node.js and npm

• Install VS Code

• Install VS Code extensions:

o Solidity

o Hardhat

• Use a testnet (Sepolia / Goerli / Mumbai)

• Create a smart contract and a script to interact with a DeFi protocol (mock lending contract)

• Use functions for deposit and withdrawal

• Test interactions using test tokens

• Add inline comments explaining the logic

Tools & Technologies

• Solidity

• Hardhat

• JavaScript (Node.js)

• Ethereum Testnet (Sepolia)

• MetaMask (Testnet account)

Practical Description

In this practical, we simulate interaction with a DeFi protocol by creating a simple lending contract where:

• Users can deposit ETH

• Users can withdraw ETH

• Contract tracks balances internally

This represents how DeFi protocols like Aave or Compound work on testnets.

CODE: